

All-Optical Contention Resolution using a single optical flip-flop and two stage all-optical wavelength conversion

C. Stamatiadis, M. Bougioukos, A. Maziotis, P. Bakopoulos, L. Stampoulidis and H. Avramopoulos

National Technical University of Athens – School of Electrical and Computer Engineering
9 Heron Polytechniou Street, Zografou 15773 – Athens, Greece
Email: cstamat@mail.ntua.gr

Abstract: We present a new scheme for 40 Gb/s all-optical contention resolution based on cascaded wavelength converters and a hybrid integrated optical flip-flop. Using simultaneously both ports of the flip-flop, component and power reduction is achieved.

©2010 Optical Society of America

OCIS codes: (130.4815) Optical switching devices; (060.6719) Switching, packet

1. Introduction

The recent advances in functional photonic integration enable the development of multi-element photonic routing systems with increased number of on-chip components, small footprint and reduced power consumption [1]. In this context the realization of complex and component-demanding routing functionalities in the optical domain, such as contention resolution, seems to be a step closer to practical implementation [2]. The requirement for small size and reduced energy per bit consumption should also affect the system design: practical systems should involve a reasonable number of active components and should be developed using a building block approach. In the case of optical contention resolution, the reported schemes employ a number of different functionalities such as Packet Envelope Detection (PED), space switching, wavelength conversion and optical latching, leading to large and complex architectures with increased number of active elements and consequently considerable power consumption [3, 4].

In this paper we demonstrate an optical circuit that performs on-the-fly contention resolution between 40 Gb/s packets of the same wavelength. It relies on a hybrid integrated optical flip-flop (OFF), a SOA-MZI gate and two wavelength converters based on the SOA-filter technique (SOA-OBF). The scheme makes optimal use of the single flip-flop by utilizing both of its outputs to perform packet-by-packet wavelength conversion either to the original (no contention) or to a new wavelength (contention). A total number of six active elements (SOAs) are adequate to resolve the contention with error-free performance at 40 Gb/s.

2. Concept

Figure 1 illustrates the schematic diagram of the contention resolution circuit consisting of an optical flip-flop and three 40 Gb/s Wavelength Converters (WCs). Packets 1 and 2 at λ_0 along with their labels at λ_3 enter the system with direction to two separate arms. The upper arm drives the packets directly to output O/P1, while the lower arm extracts the labels with narrowband optical filtering and feeds the optical flip-flop. As a consequence optical pulses at λ_0 and λ_2 are generated at the output ports of the OFF with length slightly longer than the length of a data packet. The two outputs of the optical flip-flop, being complementary, are used as input to two subsequent SOA-OBFs. Initially, packet P4 enters the system from IN2. The packet is wavelength converted to an intermediate wavelength (λ_1) and then it is launched to the two cascaded SOA-OBF WCs. Due to the absence of a packet from IN1, the OFF remains idle, emitting λ_0 from port 1 (and no light from port 2). As such SOA-OBF 1 is activated and P4 is

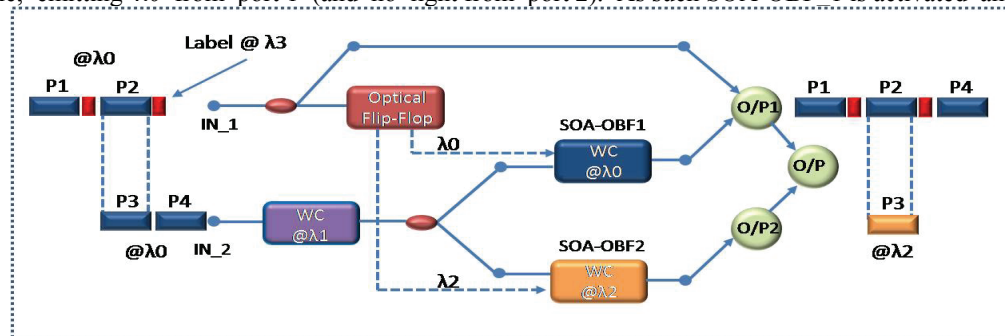


Figure 1: Contention resolution concept

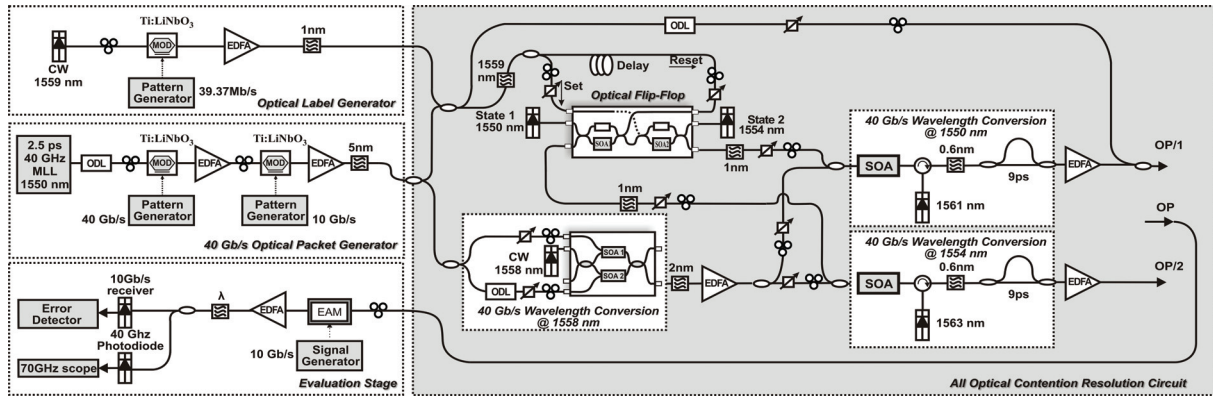


Figure 2: Experimental setup

converted to its original wavelength before leaving the system from O/P1. Subsequently packet P3 appears in IN2 on an overlapping time slot with P2 (entering from IN1). The OFF is toggled by the P2 label and provides a packet CW signal at port 2 (and no light at port 1). Thus SOA-OBF_2 is activated and P3 is converted on a new wavelength (λ_2) enabling the multiplexing of packets in a common output.

3. Experimental Setup

Figure 2 depicts the experimental setup that consists of: the 40 Gb/s optical packet generator, the optical label generator, the all-optical contention resolution unit and the evaluation stage. A 40 GHz mode-locked laser generates 2.5 ps pulse train at 1550nm that is modulated by a Li:NbO₃ modulator into 40 Gb/s data containing a PRBS 2⁷-1. This signal enters a second Li:NbO₃ modulator driven by a 10 Gb/s pattern generator in order to form a sequence of two data packets with 20.8 ns duration and 4.6 ns guard band followed by an empty packet slot. The resulting pattern is split into two parts in order to provide the two incoming packet streams delayed by 25.4 ns with respect to each other. The upper stream containing packets 1 and 2 is combined with labels of 1 ns duration generated by a third Li:NbO₃ modulator. This signal is further separated for routing the packets directly to output O/P1 and triggering the optical flip-flop. In order to provide the Set and Reset pulses, the labels are extracted using an optical filter, divided and finally delayed by 22ns. Both complementary outputs of the flip-flop are interconnected to the inputs of the two SOA-OBF WCs. The packet stream containing P3 and P4 is launched into a hybridly integrated SOA-Mach-Zehnder Interferometer (SOA-MZI) together with a Continuous Wave (CW) for wavelength conversion at 1558nm employing the push-pull technique. The switched port of SOA-MZI is separated into two parts which are introduced as control signals to the SOA-filter wavelength converters after time synchronization with the outputs of the flip-flop. Both SOAs are 1.1 mm length, driven by 300mA and counter-propagating CW assist beams were utilized in order to reduce the SOA ASE between the data packets. For chirp filtering two optical filters with 0.6 nm bandwidth were used followed by 9ps fiber delay interferometers for signal inversion. The contending packet 3 exits finally the circuit from output O/P2 at 1554nm while packet 4 wavelength converted again at 1550nm is combined with the initial data stream and exits output O/P1. The outputs of the SOA-OBF wavelength converters containing 40 Gb/s data packets were evaluated by BER measurements after demultiplexing them to 10 Gb/s using an Electro-Absorption Modulator (EAM). Table I shows the optical powers used for the three WCs.

WC	SOA-MZI	SOA-Filter (1550nm)	SOA-Filter (1554nm)
Pin	-4	2.9	4.27
Pctr	-1.6	3.76	5.14
CW counter	---	-0.4	1.7

Table I: Optical Powers for WCs in dBm

4. Results and discussion

Figure 3 shows time domain traces at different parts of the contention resolution circuit, with emphasis on input, output and contention unit signals. Figure 4 illustrates the BER performance of the input packets, O/P2 and the output of 1550 WC, along with corresponding eye-diagrams of 40Gb/s signals and 10 Gb/s demultiplexed channels. Error free operation with power penalty less than 4 dB was achieved for the output of 1550 WC (Packet 4)

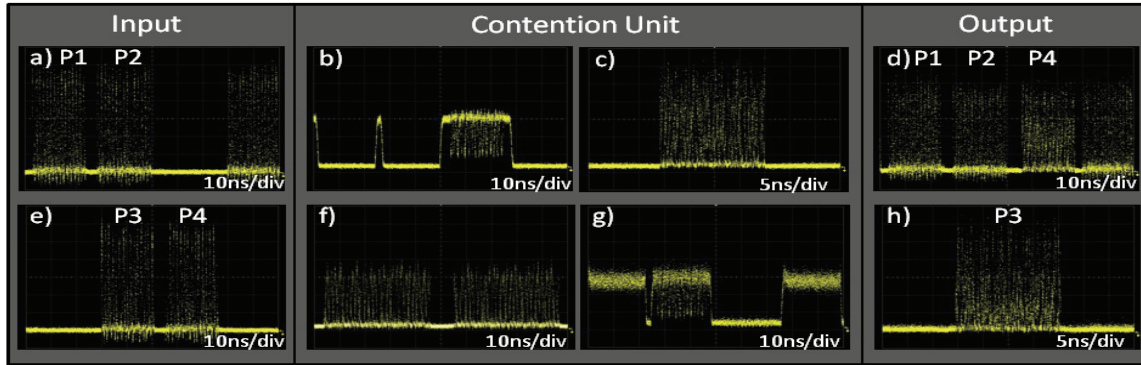


Figure 3: Traces of a) input packets 1-2, b) inverted signal at 1550 WC, c) non-inverted signal at 1550 WC, d) output of O/P1, e) input packets 3-4, f) output of SOA-MZI WC, g) inverted signal at 1554 WC, h) output of O/P2.

compared to input data packets. A BER of 10^{-8} was obtained for O/P2, which is well above the FEC limit. This is due to the fact that the OFF could not be biased perfectly resulting to different extinctions ratios at its ports [3]. The imperfect biasing stems from the not ideal coupling ratios and the lack of precise thermal heating causing unequal gain saturations and phase fluctuations. As such BER performance can be improved further with optimized fabrication of the OFF.

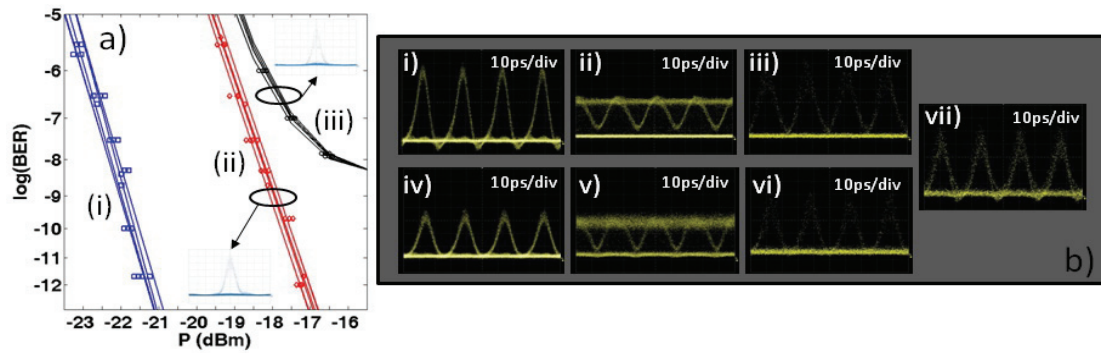


Figure 4: a) BER curves of i) input packets, ii) output of SOA-OBF_1, iii) output of SOA-OBF_2. b) 40 Gb/s eye diagrams of i) input packets, ii) inverted signal at SOA-OBF_1, iii) non-inverted signal at SOA-OBF_1, iv) output of SOA-MZI WC, v) inverted signal at 1554 WC, vi) O/P2, vii) O/P1

5. Conclusions

We have demonstrated a contention resolution circuit that employs cascaded wavelength converters connected to both ports of a single integrated optical flip-flop. The circuit operates at 40 Gb/s and can be extended at 160 Gb/s, whereas it employs a minimum total number of active components, exhibiting optimum performance with respect to electrical power consumption.

References

- [1] S. C. Nicholes, et al., "The World's First InP 8x8 Monolithic Tunable Optical Router (MOTOR) Operating at 40 Gbps Line Rate per Port", presented at the OFC/NFOEC 2009, San Diego, USA, paper PDPB1.
- [2] L. Stampoulidis et al., "Enabling Tb/s Photonic Routing: Development of Advanced Hybrid Integrated Photonic Devices to Realize High-Speed, All-Optical Packet Switching," J. Sel. Quantum Electron. **14**, 849-860 (2008)
- [3] P. Bakopoulos et al., "160 Gb/s All-Optical Contention Resolution with Prioritization Using Integrated Photonic Components," in ECOC'09, paper 6.3.5, Vienna, Austria.
- [4] P. Zakyntinos et al., "Successful Interconnection of SOA-MZI Arrays and Flip-Flops to Realize Intelligent, All-optical Routing," in ECOC'08, paper We.2.D.2, Brussels, Belgium.

Acknowledgement

This work was partially supported by European Commission through ICT-BOOM project. We gratefully acknowledge CIP Technologies for providing the hybridly integrated optical flip-flop.